CHAPTER 1

THE LIVING WORLD

CHAPTER ESSENCE AT A GLANCE:
Metabolism, cellular organization, and consciousness define life, classified through binomial nomenclature into a nested hierarchy.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Defining Properties of Living Organisms:** Consciousness, cellular organization, and metabolism represent all-inclusive defining properties of living organisms without exception. Growth and reproduction are characteristic attributes but not absolute defining criteria, as non-living crystals grow by external accretion and infertile organisms (worker bees, mules, infertile human couples) cannot reproduce.

• **Binomial Nomenclature Protocol:** Carolus Linnaeus established binomial nomenclature: each scientific name comprises two Latinized components: the generic name (capitalized) and the specific epithet (lowercase). In print, names are italicized, and when handwritten, both words are separately underlined, with author citations appended at the end (e.g., Homo sapiens Linnaeus).

• **International Taxonomic Codes:** Standardized rules for assigning unique scientific names across the globe are governed by international codes, primarily the International Code of Zoological Nomenclature (ICZN) and the International Code of Botanical Nomenclature (ICBN), ensuring that every organism has only one universally recognized scientific name.

• **Taxonomic Hierarchy Sequence:** Classification involves a nested hierarchy of obligate ascending categories: Species, Genus, Family, Order, Class, Phylum, and Kingdom. As one ascends from species to kingdom, the number of shared morphological and physiological characteristics progressively decreases, while general relatedness broadens.

ESSENTIAL PHYSIOLOGICAL PATHWAYS & STRUCTURES

$$\text{Nomenclature Format} = \textit{Homo}\ (\text{Genus, Capital}) + \textit{sapiens}\ (\text{Species, Small})\ \text{L.}$$

Binomial Nomenclature Linnaean Convention

$$\text{Taxonomic Hierarchy} = \text{Species} \rightarrow \text{Genus} \rightarrow \text{Family} \rightarrow \text{Order} \rightarrow \text{Class} \rightarrow \text{Phylum} \rightarrow \text{Kingdom}$$

Obligate Ascending Taxonomic Categories

$$\text{Human Lineage} = \text{Hominidae (Family)} \rightarrow \text{Primates (Order)} \rightarrow \text{Mammalia (Class)} \rightarrow \text{Chordata (Phylum)}$$

Taxonomic Lineage of Modern Humans

$$\text{Life Attributes} = \frac{\text{Defining: Metabolism, Cellular Organization, Consciousness}}{\text{Non-Defining: Extrinsic Growth, Reproduction}}$$

Classification Criteria of Biological Entities

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Remember that Taxonomic Aids (Herbarium, Botanical Gardens, Museum, Zoological Parks, and Keys) were completely deleted under the rationalized syllabus.

• **Alert:** In binomial nomenclature, the generic name always begins with a capital letter, while the specific epithet begins with a small letter (e.g., Panthera leo).

• **Alert:** Metabolism is a defining feature of all living organisms without exception, whereas growth is not defining because non-living objects can accumulate material externally.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Why are reproduction and growth not considered defining characteristics of living organisms? [Score: 2 Marks]

Answer & Key Valuation Steps:

• Growth: Non-living objects like mountains, boulders, and sand dunes also increase in mass by extrinsic accumulation of material on their surface. **[1 Mark]**

• Reproduction: Many living organisms do not reproduce at all, such as sterile worker bees, mules, and infertile human couples, yet they are alive. **[1 Mark]**

2. State any four universal rules of binomial nomenclature formulated by Carolus Linnaeus. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Biological names are generally in Latin and written in italics; they are Latinized or derived from Latin irrespective of origin. **[0.5 Mark]**

• The first word represents the genus, while the second component denotes the specific epithet. **[0.5 Mark]**

• Both words when handwritten are separately underlined, or printed in italics to indicate their Latin origin. **[0.5 Mark]**

• The genus name begins with a capital letter, while the specific epithet starts with a small letter (e.g., Mangifera indica). **[0.5 Mark]**

3. Explain why cellular organization, metabolism, and consciousness are considered definitive properties of all living beings. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Cellular organization: All living organisms consist of one or more cells; life cannot exist below the level of complete cellular architecture. **[1 Mark]**

• Metabolism: The sum total of all anabolic and catabolic chemical reactions occurring in the body; metabolic reactions occur in all living forms without exception. **[1 Mark]**

• Consciousness: The ability of living organisms to sense their physical, chemical, or biological surroundings and respond appropriately to external environmental stimuli. **[1 Mark]**

4. Arrange the following taxonomic categories in proper ascending sequence and assign the correct category for modern humans (Homo sapiens): Order, Phylum, Genus, Family, Class, Species. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Ascending sequence: Species -> Genus -> Family -> Order -> Class -> Phylum. **[1.5 Marks]**

• Classification for Homo sapiens: Species: sapiens; Genus: Homo; Family: Hominidae; Order: Primata; Class: Mammalia; Phylum: Chordata. **[1.5 Marks]**

5. What happens to the number of common characteristics shared among organisms as one moves from species to kingdom in a taxonomic hierarchy? Explain with an example. [Score: 3 Marks]

Answer & Key Valuation Steps:

• As one goes higher from species to kingdom, the number of common characteristics shared between organisms goes on decreasing. **[1 Mark]**

• Lower taxa (species) share maximum fundamental morphological and reproductive attributes, allowing interbreeding. **[1 Mark]**

• Higher categories (Phylum, Kingdom) share only very broad generalized organizational plans (e.g., presence of notochord in all chordates from fish to mammals). **[1 Mark]**

6. Identify the scientific binomial names and families of: (a) Housefly, (b) Lion, and (c) Modern human. [Score: 3 Marks]

Answer & Key Valuation Steps:

• (a) Housefly: Scientific name: Musca domestica; Family: Muscidae. **[1 Mark]**

• (b) Lion: Scientific name: Panthera leo; Family: Felidae. **[1 Mark]**

• (c) Modern human: Scientific name: Homo sapiens; Family: Hominidae. **[1 Mark]**

7. A crystal in a chemical solution grows in size, while an infertile worker honeybee and a mule cannot reproduce. Justify why the crystal is classified as non-living while the sterile mule is classified as living. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• Crystal growth is non-living: Crystal growth is extrinsic (accretion), occurring purely by physical deposition of dissolved solute particles onto its outer surface without internal metabolism or cellular division. **[1.5 Marks]**

• Mule sterility is living: Although infertile due to an odd chromosome number from interspecific hybridization (horse and donkey cross), the mule exhibits active cellular organization, respiration, and thousands of continuous metabolic reactions. **[1.5 Marks]**

• Conclusion: Intrinsic cellular metabolism and environmental consciousness define life; reproductive capability is secondary. **[1 Mark]**